# Analysis of Unemployment Trends in India

*Impact of the COVID-19 Pandemic on Regional Employment (2020)*

Task Report

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# 1. Introduction

The COVID-19 pandemic that began in early 2020 was one of the biggest economic disruptions in recent history. When India entered a nationwide lockdown in late March 2020, millions of workers suddenly lost their jobs or had their income sharply reduced. This was especially hard on daily wage workers, small business owners, and people in the informal economy who had no savings or job security to fall back on.

This report analyses how unemployment in India changed throughout 2020. It looks at which regions were hit the hardest, how the situation compared between urban and rural areas, whether the spike in unemployment during the lockdown was statistically significant, and which states showed similar patterns over time. The aim is to go beyond just describing what happened and to use data analysis techniques to find patterns and insights that are not immediately obvious.

### 1.1 Datasets Used

Two datasets were used in this project:

| Dataset | Coverage | Key Columns |
|---|---|---|
| Unemployment_Rate_upto_11_2020.csv | January – November 2020, 27 Indian states | Region, Date, Unemployment Rate (%), Estimated Employed, Labour Participation Rate (%) |
| Unemployment in India.csv | 2019 onwards, Urban & Rural split | Region, Date, Unemployment Rate (%), Area (Urban / Rural) |

The first dataset is the main one used for trend analysis, statistical testing, SNA, and the animated map. The second dataset was used specifically for the urban vs. rural comparison.

# 2. Methodology

The analysis was done entirely in Python. The following tools and techniques were used:

| Step | Technique | Tool / Library | Purpose |
|---|---|---|---|
| 1 | Time-series line chart | matplotlib, seaborn | Visualise unemployment over time |
| 2 | Boxplot comparison | seaborn | Compare spread between urban and rural areas |

| | | | |
|---|---|---|---|
| 3 | Two-sample t-test | scipy.stats | Test if lockdown caused a significant change |
| 4 | Social Network Analysis | networkx | Find regions with similar patterns |
| 5 | Animated geospatial map | plotly | Show month-by-month geographic spread |

Before running the analysis, the data was cleaned and prepared. Date columns were converted to proper datetime format, and month and year fields were extracted. One important fix was correcting a column swap in the CSV file, where the latitude and longitude values had been stored in the wrong columns. If left uncorrected, this would have placed every data point in the wrong location on the map.

For the SNA part, a correlation matrix was built between all regions based on their monthly unemployment rates. If two regions had a Pearson correlation above 0.80, an edge was drawn between them in the network. Betweenness centrality was then calculated to find which region acted as the most important connector between different parts of the network.

## 3. Results and Discussion

### 3.1 Unemployment Rate Trend Over Time

Figure 1 shows how the estimated national unemployment rate changed month by month throughout 2020. The shaded red region marks the initial lockdown period, which ran from late March to the end of June 2020.

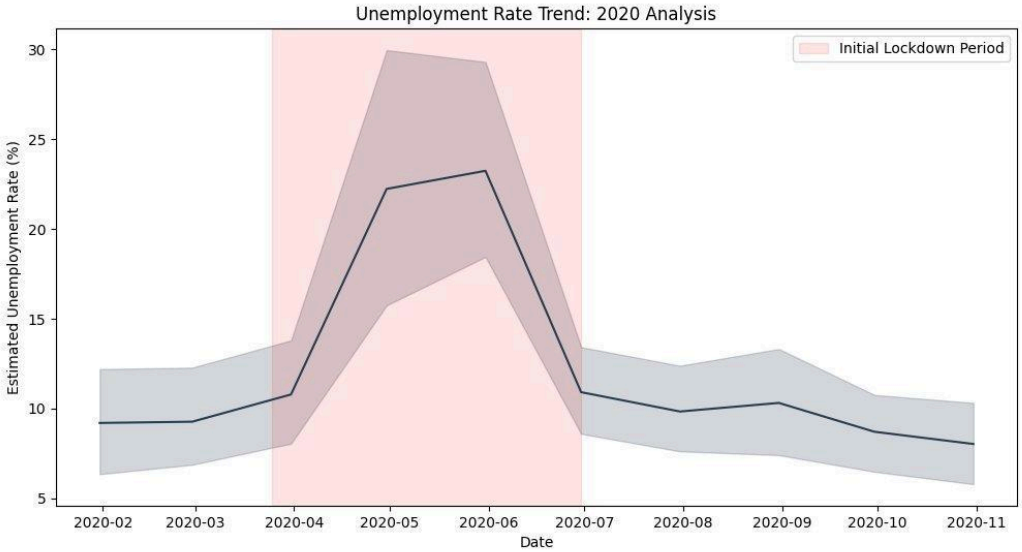


*Figure 1: Monthly Unemployment Rate Trend (Jan – Nov 2020) with Lockdown Period Highlighted*

Before the lockdown, unemployment was relatively stable at around 9%. Once restrictions came into force in April, the rate jumped sharply, peaking at around 23% in May 2020. This is more than double the pre-lockdown baseline and represents a massive loss of income for a huge number of people.

After the lockdown ended in late June, the rate dropped back quickly. By July it was already close to 10%, and it continued to decline slowly through the rest of the year. By October and November it had settled at around 8 to 9%, which is actually slightly below where it started. This gives the overall trend a V-shaped look, but as the following sections show, this average hides a lot of regional variation.

The confidence band (the shaded grey area around the line) is notably wider during the lockdown months. This tells us that while the national average was rising, different states were being affected to very different degrees – some saw unemployment go up to 70% or more, while others were far less impacted.

### 3.2 Urban vs. Rural Unemployment Distribution

Figure 2 compares the spread of unemployment rates between urban and rural areas. The boxplot shows the median, the interquartile range (the box), the full range (the whiskers), and any outliers beyond that.

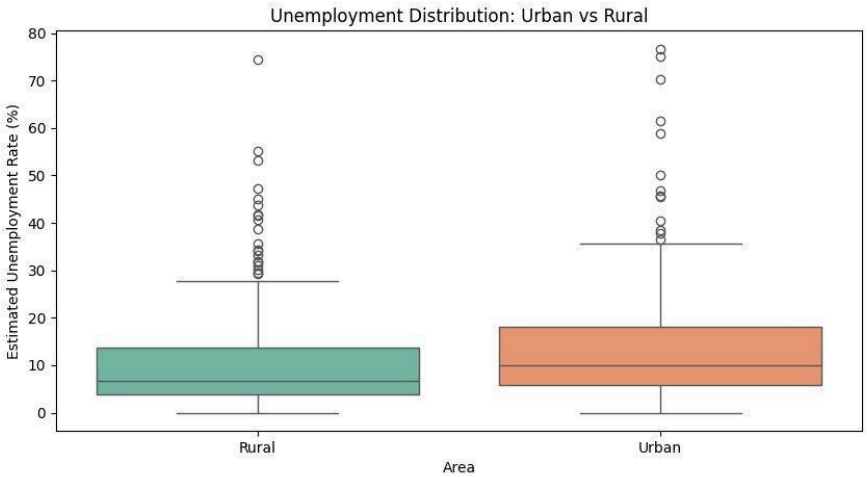


*Figure 2: Unemployment Distribution – Urban vs. Rural Areas*

At first glance, both distributions look fairly similar as the medians are close to each other at around 9 to 10%. But looking more carefully, urban areas have a noticeably wider box, which means there is much more variation in urban unemployment rates. Some urban regions had very low unemployment, while others had extremely high rates.

Both groups have outliers in the 55 to 75% range. These are almost certainly from the peak lockdown months in specific states, where entire industries shut down overnight. The rural distribution being slightly tighter makes sense, because rural economies in India are heavily dependent on agriculture, which continued functioning during the lockdown to a greater degree than most other sectors.

This comparison highlights an important point: simple national averages hide the fact that urban workers, who are more likely to be employed in services, retail, and construction, were exposed to a very different kind of risk than rural workers.

### 3.3 Statistical Analysis: Lockdown vs. Recovery Period

To go beyond a visual comparison, a two-sample t-test was run to check whether the unemployment rate during the lockdown period (April to June 2020) was significantly different from the rate during the recovery period (July to November 2020). Figure 3 shows the mean values for both periods.

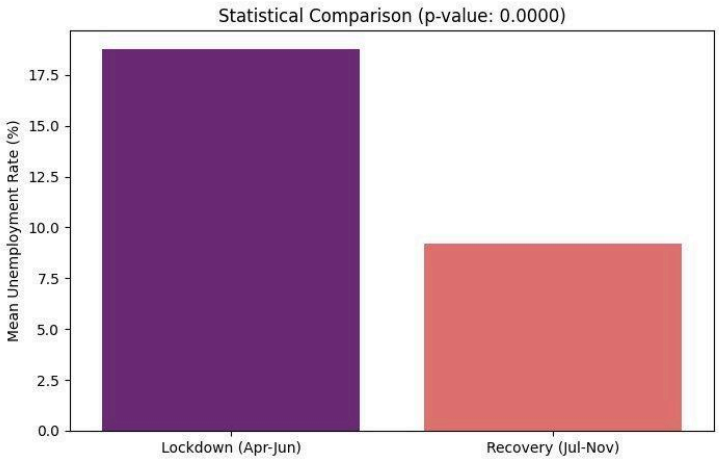


*Figure 3: Mean Unemployment Rate – Lockdown Period vs. Recovery Period*

| Metric | Value |
|---|---|
| Mean Unemployment – Lockdown (Apr–Jun 2020) | ~18.7% |
| Mean Unemployment – Recovery (Jul–Nov 2020) | ~9.3% |
| T-Statistic | 5.9586 |
| P-Value | 0.0000 (< 0.001) |

The p-value of essentially zero confirms that the difference between these two periods is statistically significant, in other words, the lockdown did cause a real and measurable spike in unemployment, not just random variation in the data. The t-statistic of 5.96 is high enough that there is no ambiguity here that the two periods are fundamentally different in terms of their unemployment levels.

To put the numbers in simple terms, during the lockdown, the average unemployment rate was almost exactly double what it was during the recovery phase. The recovery was meaningful and real, but it is worth noting that even the recovery-phase average of 9.3% is higher than the pre-pandemic baseline, suggesting the economy had not fully returned to normal by the end of 2020.

### 3.4 Social Network Analysis – Regional Correlations

Social Network Analysis (SNA) is a method for understanding relationships and connections. It is most commonly used for social media or communication data, but it can be applied to any situation where

you want to understand how different entities are related to each other. In this case, it was used to find out which Indian states experienced similar unemployment patterns over the course of 2020.

The logic is such that if two states had a Pearson correlation of more than 0.80 in their monthly unemployment rates, they were connected by an edge in the network graph. A high correlation means both states went up and came down at roughly the same time and by similar amounts. Figure 4 shows the resulting network.

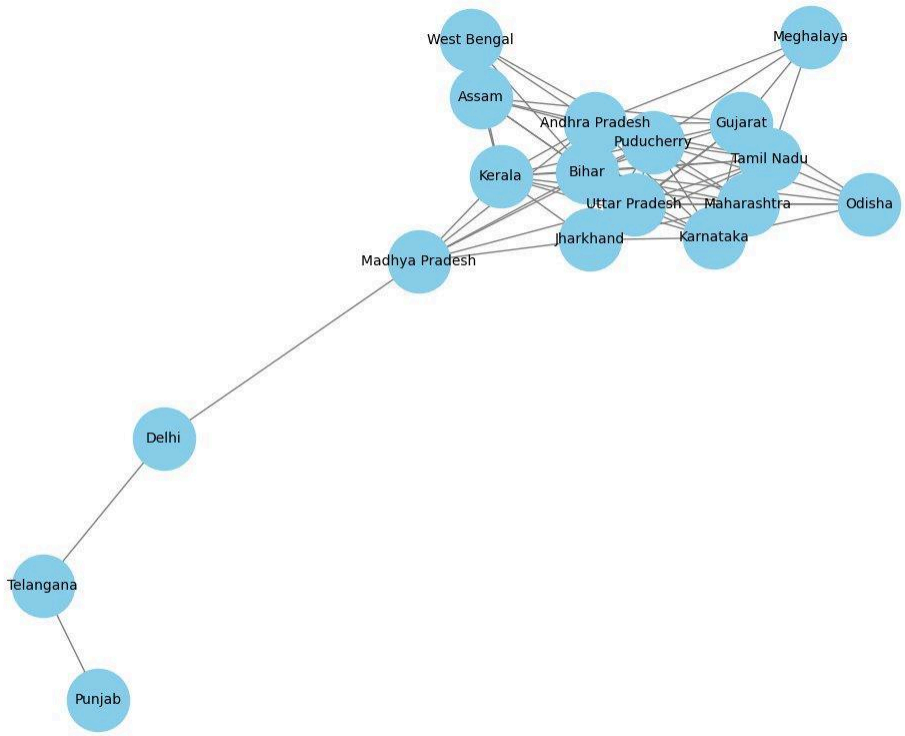


*Figure 4: Regional Unemployment Correlation Network (Edge = Correlation > 0.80)*

The network clearly shows two main clusters. The large, dense cluster on the right side includes over 15 states: Andhra Pradesh, Gujarat, Tamil Nadu, Maharashtra, Kerala, Bihar, Uttar Pradesh, Jharkhand, Karnataka, Odisha, Puducherry, and others. These states had very similar unemployment trajectories, likely because they are all large states with diverse economies that went through the same cycle of shutdown and reopening.

The smaller cluster at the bottom-left contains Delhi, Punjab, and Telangana. These three states appear to have followed a somewhat different path. Delhi, as the national capital with a high concentration of

government and service sector jobs, probably had a different experience from a manufacturing-heavy state. Telangana is a major IT hub, and a large part of its workforce was able to shift to remote work. Punjab's economy is heavily agricultural, which also sets it apart from most other states.

The most significant finding from the SNA is that Madhya Pradesh had the highest betweenness centrality score, making it the top "bridge" region in the network. Betweenness centrality measures how often a node sits on the shortest path between two other nodes. A high score means that Madhya Pradesh connected the two clusters, acting as a relay point between them. This suggests that whatever happened to employment in Madhya Pradesh had implications for both groups of states, making it a strategically important region for economic policy decisions.

## 4. Summary of Key Findings

| Finding | Detail |
|---|---|
| Peak Unemployment Month | May 2020 – national average reached approximately 23% |
| Speed of Recovery | Dropped back to ~10% by July 2020, just one month after restrictions eased |
| Statistical Test Result | T-stat = 5.96, p-value ≈ 0.0000 – lockdown impact is statistically significant |
| Urban vs. Rural | Urban areas showed higher variability; rural areas more stable but still severely affected during peak |
| SNA Bridge Region | Madhya Pradesh – highest betweenness centrality, connects two regional clusters |
| Main Cluster | 15+ states with high unemployment correlation ($r > 0.80$), similar economic trajectories |
| Separate Cluster | Delhi, Punjab, Telangana – distinct patterns due to sector composition (IT, agriculture, government) |

## 5. Limitations

Every analysis has limitations, and it is important to be honest about them.

The data only goes up to November 2020, which means the second wave of COVID-19 in early 2021, which was even more severe in terms of health impact, is not captured here. The recovery seen in the second half of 2020 may have been followed by another sharp increase.

The dataset works with state-level monthly averages. This hides a lot of within-state variation. For example, Maharashtra as a whole might show a moderate average, but within it, Mumbai and smaller rural districts would have had very different experiences.

There is also no breakdown by gender, age group, or employment type. Research has shown that women, young people, and informal workers were disproportionately affected by pandemic-related job losses. A more detailed dataset would allow a much richer analysis of who was most impacted.

Finally, the SNA correlation threshold of 0.80 is somewhat arbitrary. A different threshold would produce a different network and potentially a different bridge region. Future work could test multiple thresholds and examine how robust the results are.

## 6. Conclusion

This project set out to understand how the COVID-19 pandemic affected unemployment across India in 2020. Using a combination of visualisation, statistical testing, and network analysis, it is clear that the lockdown had a dramatic and measurable impact on employment, causing the national average unemployment rate to more than double in just a few weeks.

The statistical test confirmed that this was not a random fluctuation – the difference between the lockdown and recovery periods is highly significant. The urban vs. rural comparison showed that while both groups were affected, urban areas experienced more extreme volatility. And the Social Network Analysis revealed that most Indian states shared similar unemployment trajectories, with Madhya Pradesh acting as a key bridge between different regional groups.

What makes this analysis interesting from a computer science perspective is the combination of different techniques applied to the same dataset. Each method on its own tells part of the story. Together, they give a much more complete and nuanced picture than any single chart or number could.

Going forward, a more detailed dataset with demographic breakdowns, district-level data, and coverage through 2021 would allow for a much richer analysis. It would also be interesting to build a predictive model to estimate unemployment based on lockdown stringency, sector composition, and other factors.

## 7. References

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